



Source: <https://github.com/mav8557/Father>

Linux Forensic Scenario Report



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Note: this report was rewritten from the original PDF submission.

Overview

The alert is for unencrypted traffic on port 22/tcp, specifically the string "python3 -c 'import pty; pty.spawn("/bin/bash")'" which triggered the alert for "reverse shell promotion" in the NMS. The system is showing signs of heavy CPU usage.

System parameters

- Hostname - vbox.fci.int
- OS - Debian 13
- IP address - 192.168.4.22
- Timezone - UTC-4

All timestamps below are in the UTC timezone.

Conclusions

The Father rootkit, which persisted via /etc/ld.so.preload, was deployed no later than at 2026-03-24 23:16:29. The initial method of rootkit deployment remains unidentified. Leveraging the rootkit, the attacker obtained a root shell after connecting from 192.168.4.35:48411 at 23:26:05 via "nc" connection. Subsequently, the attacker established an SSH connection to the system **192.168.5.95** using the **pocmp** account and the parameter "ymv", and deployed the XMRig miner on vbox system, renaming it to "top".

Investigation

The process PID 937 "ssh" received the incoming connection from 192.168.4.35:48411. The port 48411 corresponds the rootkit Father located at /usr/lib/x86_64-linux-gnu/libymv.so.3 (MD5: e2c2e24aa54c08b722a03f39b78c95dc), so the rootkit was activated

after the connection. The functional description of the rootkit is given in the "Malware analysis: Rootkit libymv.so.3" section.

Using the given shell, the attacker connected since 2026-03-24 23:26:05 (/dev/pts/2 ctime) until 2026-03-24 23:29:36 (/dev/pts/2 mtime, atime). The attacker used **worker** account and elevated his privileges to **root** via the rootkit.

The child processes of PID 937, PIDs 939, 940, 941, 975, 977, are hidden. The pstree fragment from memory is listed below:

```
* 0x8c7cc67e1980      937      937      1      sshd
** 0x8c7cc8278000     939      939     937      sh
*** 0x8c7cc67e4c80    940      940     939
python3
**** 0x8c7cc6011980   941      941     940      bash
***** 0x8c7cc6014c80 975      975     941      ssh
```

The detailed description of the processes:

PID	Start time	Exe	Commandline	Comment
937	2026-03-24 23:25:37	/usr/sbin/sshd	sshd: /usr/sbin/ss	SSH process which received the incoming connection from 192.168.4.35:48411
939	2026-03-24 23:26:07	/bin/sh	/bin/sh	The process which is programmatically started by the rootkit in accept() hook
940	2026-03-24 23:26:22	/usr/bin/python3.13	python3 -c 'import pty; pty.spawn("/bin/bash")'	The custom command entered by the attacker
941	2026-03-24 23:26:22	/usr/bin/bash	/bin/bash	The shell where in-memory commands were executed
975	2026-03-24 23:28:32	/usr/bin/ssh	ssh -F config ymv	Connection to 192.168.5.95:22 using pocmp@192.168.5.95
977	2026-03-24 23:29:24	/dev/shm/kit/top	top -o 127.0.0.1:3333 -B	xmrig

Having got the shell, the attacked deployed in the working directory /dev/shm/ xmrig miner, renamed it to "top" and started with the command server 127.0.0.1:3333, then removed the file. Here is the bash history for PID 941 from memory, parsed via Volatility:

```
941      bash      2026-03-24 23:26:26.000000 UTC      rest
941      bash      2026-03-24 23:26:30.000000 UTC      reset
941      bash      2026-03-24 23:26:44.000000 UTC      echo
$SHELL
941      bash      2026-03-24 23:26:54.000000 UTC      id
941      bash      2026-03-24 23:27:25.000000 UTC      cd
/dev/shm/kit
941      bash      2026-03-24 23:27:26.000000 UTC      ls
941      bash      2026-03-24 23:27:51.000000 UTC      mv
xmrig top
941      bash      2026-03-24 23:27:59.000000 UTC
export PATH=.:$PATH
941      bash      2026-03-24 23:28:17.000000 UTC      ssh -
F config ymv
941      bash      2026-03-24 23:28:28.000000 UTC      bg
941      bash      2026-03-24 23:29:09.000000 UTC      top -
o 127.0.0.1:3333 -B
941      bash      2026-03-24 23:29:15.000000 UTC      cd ..
```

```

941      bash      2026-03-24 23:29:19.000000 UTC  rm -
rf kit
941      bash      2026-03-24 23:29:24.000000 UTC  ps -
ef | grep top
941      bash      2026-03-24 23:29:36.000000 UTC  ls

```

Additionally, there are the commands with their output from the memory dump:

```

root@vbox:/# rest
bash: rest: command not found
root@vbox:/# reset
reset: unknown terminal type unknown
Terminal type? xterm
root@vbox:/# export
echo $SHELL
/bin/bash
root@vbox:/# id
uid=0(root) gid=7823 groups=7823
root@vbox:/# cd /dev/shm/kit
root@vbox:/dev/shm/kit# ls
config rk.so xmrig
root@vbox:/dev/shm/kit# mv xmrig top
root@vbox:/dev/shm/kit# export PATH=.:$PATH
root@vbox:/dev/shm/kit# ssh -F config ymv
Warning: Permanently added '192.168.5.95' (ED25519)
to the list of known hosts.
pocmp@192.168.5.95's password:
[1]+  Stopped                  ssh -F config ymv
root@vbox:/dev/shm/kit# bg
[1]+  ssh -F config ymv &
root@vbox:/dev/shm/kit# top -o 127.0.0.1:3333 -B
root@vbox:/dev/shm/kit# ps -ef |
cd ..
root@vbox:/dev/shm# rm -rf kit
root@vbox:/dev/shm# ps -ef | grep top
root      977      1 99 19:29 ?        00:00:53
top -o 127.0.0.1:3333 -B
root      993      941  0 19:29 pts/2    00:00:00
grep top
root@vbox:/dev/shm# ls

```

GID **7823** corresponds the magic value from the rootkit config. The content of "config" file was not identified, but it contained parameters for a connection to IP **192.168.5.95**. The command "ssh -F config ymv" probably means "connect to a host with the parameters from config file using ymv hostname". "rk.so" is the compiled Father rootkit library. The traces of using it were not found in the given data, and it appeared later than the rootkit was actually deployed.

Due to the memory parsing result, there is still the active connection to the host **192.168.5.95**:

```

ssh 975 975 3 0x8c7cc405cc00 AF_INET STREAM TCP
192.168.4.22 33440 192.168.5.95 22 ESTABLISHED -

```

It is discovered that xmrigr uses the 127.0.0.1:3333 C2 server. There is the connection in the network output and the string in memory which states that packets from 3333 port are redirected to 59182 port and then possibly to the remote host. The directive "direct-tcpip" refers to a local port forwarding mechanism (-L) where the SSH client listens on a local port and tunnels incoming connections to a specified target host and port via the SSH server.

```
direct-tcpip: listening port 3333 for 127.0.0.1 port
3333, connect from 127.0.0.1 port 59182 to 127.0.0.1
port 3333
```

The file /tmp/silly.txt (all timestamps are 2026-03-24 23:34:32) contains credentials for **worker**, hooked by the rootkit. The timestamps correspond to the last **worker** login at 23:34:32, according to /var/log/wtmp and Systemd journal logs.

Despite the timestamps of /etc/ld.so.preload (mtime 2026-03-24 23:25:19 btime: 2026-03-24 23:25:09) and /usr/lib/x86_64-linux-gnu/libbmv.so.3 (mtime, ctime, btime 2026-03-24 23:24:51, atime 2026-03-24 23:25:19), the rootkit was deployed no later than 23:16:29 (system reboot). There are earlier traces of presence of the file with the same name in different logs:

- /var/log/journal
- /var/log/boot.log
- /var/log/cups/error_log

There are errors in the logs, which witness about existence of the file at the same path as the rootkit no later than 23:16:30, when its execution failed.

```
----- Tue Mar 24 19:16:30 EDT 2026 -----
-
/dev/sda1: recovering journal
/dev/sda1: clean, 160725/1240320 files,
3573749/4960256 blocks
ERROR: ld.so: object '/lib/x86_64-linux-
gnu/libbmv.so.3' from /etc/ld.so.preload cannot be
preloaded (file too short): ignored.
```

According to Systemd journal logs, an attacker connected from **192.168.4.35** during 24.03.2026 23:23:34-23:23:48.

```
Mar 24 23:23:34 vbox sudo[910]: worker : TTY=pts/0
; PWD=/home/worker ; USER=root ; COMMAND=/bin/bash
Mar 24 23:23:34 vbox sudo[910]:
pam_unix(sudo:session): session opened for user
root(uid=0) by worker(uid=1000)
Mar 24 23:23:48 vbox sshd-session[914]: Accepted
password for worker from 192.168.4.35 port 55504 ssh2
Mar 24 23:23:48 vbox sshd-session[914]:
pam_unix(sshd:session): session opened for user
worker(uid=1000) by worker(uid=0)
Mar 24 23:23:48 vbox systemd-logind[710]: New session
```

```
3 of user worker.
Mar 24 23:23:48 vbox systemd[1]: Started session-
3.scope - Session 3 of User worker.
Mar 24 23:23:48 vbox sshd-session[923]: Received
disconnect from 192.168.4.35 port 55504:11:
disconnected by user
Mar 24 23:23:48 vbox sshd-session[923]: Disconnected
from user worker 192.168.4.35 port 55504
Mar 24 23:23:48 vbox sshd-session[914]:
pam_unix(sshd:session): session closed for user
worker
```

Atime of `"/run/shm -> /dev/shm"` is 2026-03-24 23:23:48 and matches the logout time, so it's assumed that this SSH session was related to the attacker.

There is the logout error `"syslogin_perform_logout: logout() returned an error"` for the session of **worker** from **192.168.4.35** at 23:27:16 in Systemd journal logs, which indicates missing of `/run/utmp` file. No reliable connection to the rootkit was found.

```
Mar 24 23:22:19 vbox sshd-session[834]: Accepted
password for worker from 192.168.4.35 port 48364 ssh2
Mar 24 23:22:19 vbox sshd-session[834]:
pam_unix(sshd:session): session opened for user
worker(uid=1000) by worker(uid=0)
<...>
Mar 24 23:27:16 vbox sshd-session[834]:
syslogin_perform_logout: logout() returned an error
Mar 24 23:27:16 vbox sshd-session[873]: Received
disconnect from 192.168.4.35 port 48364:11:
disconnected by user
Mar 24 23:27:16 vbox sshd-session[873]: Disconnected
from user worker 192.168.4.35 port 48364
Mar 24 23:27:16 vbox sshd-session[834]:
pam_unix(sshd:session): session closed for user
worker
```

Malware analysis

Path: `/usr/lib/x86_64-linux-gnu/libymv.so.3`

MD5: `e2c2e24aa54c08b722a03f39b78c95dc`

The sample is the userland rootkit **Father** (<https://github.com/mav8557/Father>).

The functionality:

- To give a root shell if `SHELL_PASS` is provided and an SSH connection is established from the `SOURCEPORT`
- To hide `/etc/ld.so.preload` and related files with a magic GID
- To hide network connections via `/proc/net/tcp` output modification
- To hook passwords via PAM and write them to a log `/tmp/silly.txt`
- To break GnuPG signature verification

The config, corresponding to `father.h`:

- GID 7823
- SOURCEPORT 48411
- ENV ymv
- STRING isn't used, isn't defined
- PRELOAD /usr/lib/x86_64-linux-gnu/libymv.so.3
- HIDDENPORT D431 (default)
- SHELL_PASS ymv

The shell on a target is spawned by the command "ncat \$IP 22 -p \$SOURCEPORT" on a client machine.

Indicators

Files

/usr/lib/x86_64-linux-gnu/libymv.so.3 - Father
rootkit <https://github.com/mav8557/Father>

MD5 e2c2e24aa54c08b722a03f39b78c95dc

SHA1 0fd709f09c073df274e272aabca3e0f3487f9e

/dev/shm/kit/top (memory-only) - XMRig miner

Accounts

pocmp – account on 192.168.5.95 system

Other

/tmp/silly.txt – contains hooked passwords

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7h ...

Congrats on the co-win and the write-up, well done. It's a great reminder of how far curiosity and continuous learning can take you. Best of luck in your future endeavours!

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